Supplementary material: Camera-Motion Compensation for DeepSORT Person Tracking — Benefit and Cost Across Camera Motions

Table S1: MOT17 training sequences used (camera, viewpoint and conditions from the MOT16 sequence table [MOT16]; ground-truth statistics from the official annotations, pedestrian class only; detections: YOLOv8n, this study)

Seq.	Camera	Resolution	FPS	Frames	View, conditions	GT boxes	IDs	Boxes/frame	Vis. < 0.5 (%)	Med. height (px)	Dets.
02	static	1920×1080	30	600	medium, cloudy	18,581	62	31.0	60.8	102	6,911
04	static	1920×1080	30	1,050	high, night	47,557	83	45.3	39.2	171	24,492
09	static	1920×1080	30	525	low, indoor	5,325	26	10.1	40.8	285	5,748
05	moving	640×480	14	837	medium, sunny	6,917	133	8.3	47.0	189	6,474
10	moving	1920×1080	30	654	medium, night	12,839	57	19.6	28.6	115	7,588
11	moving	1920×1080	30	900	medium, indoor	9,436	75	10.5	35.0	359	7,608
13	moving	1920×1080	25	750	high, sunny	11,642	110	15.5	30.9	84	5,387

Table S2: Estimation error against the true synthetic offset (px, full resolution, three sequences, non-fallback frames; second run) and IDF1 gap to the oracle (estimated − oracle, combined). Aff.: affine; Tr.: translation-only.

	Mean error			95th pct.			IDF1 gap		
Level	Aff.	Tr.	ECC	Aff.	Tr.	ECC	Aff. (1)	Aff. (2)	ECC (2)
pan 2	0.13	0.09	0.83	0.4	0.3	2.8	+0.02	−0.47	−0.07
pan 5	0.11	0.07	0.75	0.3	0.2	2.4	0.00	−0.01	+0.33
pan 10	0.13	0.09	0.87	0.4	0.3	2.9	−0.03	0.00	−0.01
pan 20	0.14	0.09	0.88	0.4	0.3	3.1	0.00	−0.01	+0.02
shake 2	0.12	0.07	0.83	0.3	0.3	2.9	−0.07	+0.01	−0.12
shake 5	0.12	0.07	0.82	0.3	0.3	2.9	0.00	+0.18	+0.12
shake 10	0.13	0.07	0.83	0.3	0.3	2.8	+0.38	0.00	+0.03

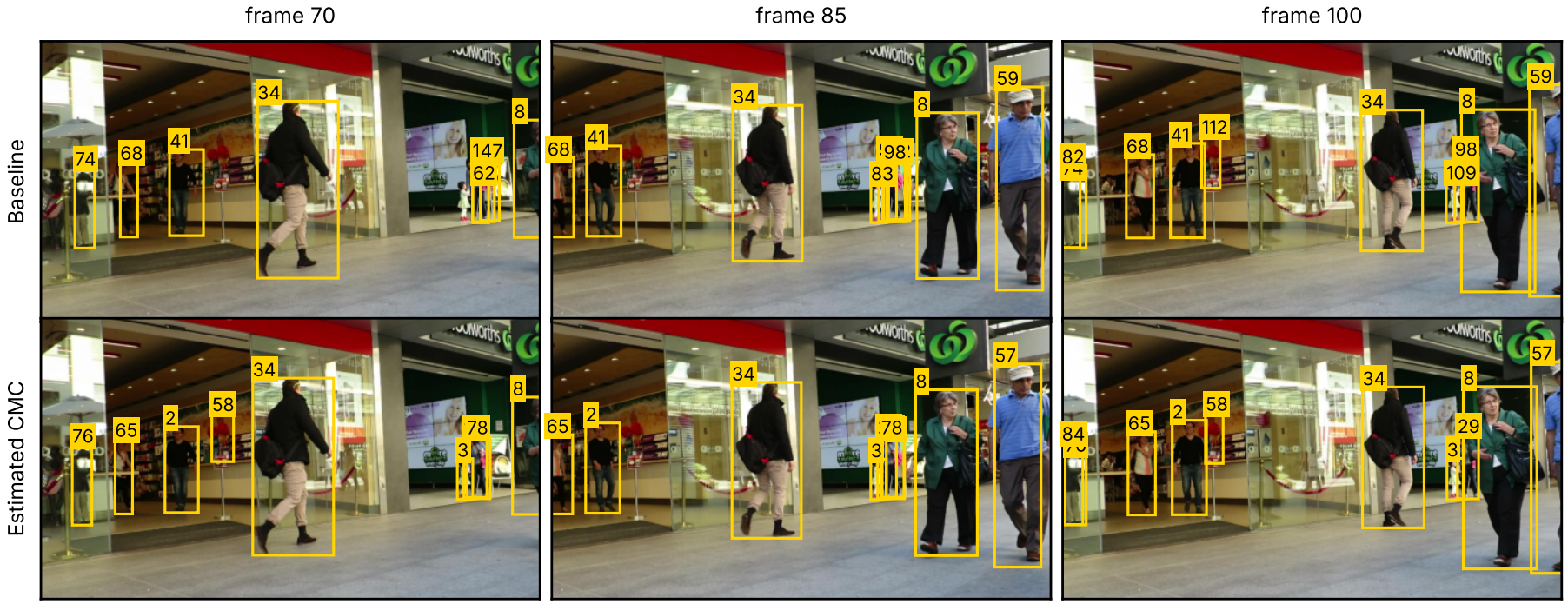


Figure S1: MOT17-09 with a synthetic 20 px/frame pan, frames 70, 85 and 100 (run 1). Top: baseline; bottom: estimated CMC, same detections. Beside the shop window, baseline identities change (47, 62; 83, 98; 109, 112), while CMC keeps identity 3 throughout. Frames from MOT17 [MOT16] (CC BY-NC-SA 3.0); overlays produced by the authors.

Table S3: Real MOT17 sequences with YOLOv8n detections: baseline scores and change with affine CMC (Δ = CMC − baseline). Sev.: median inter-frame camera motion, % of image width. Combined rows are TrackEval combined scores. Frag: track fragmentations.

	Sev.	HOTA		AssA		IDF1		IDSW		DetA		MOTA		Frag	
Sequence	(%)	Base	Δ	Base	Δ	Base	Δ	Base	Δ	Base	Δ	Base	Δ	Base	Δ
02 (static)	0.01	21.3	0.00	25.0	+0.01	23.2	+0.01	96	+1	18.4	0.00	13.1	+0.01	123	0
04 (static)	0.00	36.5	−0.02	40.0	−0.04	42.6	−0.05	187	+1	33.6	0.00	33.5	+0.01	367	+1
09 (static)	0.01	40.1	0.00	34.6	0.00	47.0	0.00	104	0	46.9	+0.01	44.2	+0.02	83	0
Static (comb.)		33.5	−0.01	37.1	−0.03	38.5	−0.03	387	+2	30.7	0.00	29.0	+0.01	573	+1
05 (moving)	0.62	43.4	+0.70	46.4	+0.95	59.7	+0.45	113	−18	40.9	+0.45	39.2	+1.00	142	−5
10 (moving)	0.20	32.0	+0.29	32.5	+0.25	37.8	−0.03	137	+2	31.8	+0.33	32.5	+0.18	226	+1
11 (moving)	0.23	47.3	+0.29	47.6	+0.38	52.8	+0.03	45	+1	47.3	+0.20	47.4	+0.02	46	+1
13 (moving)	0.37	27.8	+0.81	34.2	+1.56	35.6	+0.68	146	−10	23.5	+0.17	21.3	−0.04	166	+4
Moving (comb.)		37.5	+0.48	41.2	+0.69	45.3	+0.25	441	−25	34.7	+0.27	33.9	+0.22	580	+1
All (comb.)		35.1	+0.18	38.8	+0.26	41.1	+0.08	828	−23	32.2	+0.10	30.8	+0.09	1153	+2

Table S4: Estimators on real sequences, two detectors (Δ = condition − baseline, TrackEval combined for groups). Aff.: affine; Tr.: translation-only; Hyb.: affine with ECC on affine-fallback frames (exploratory). CI: 95% paired block-bootstrap interval of ΔHOTA (1,000 resamples; moving sequences only). FB: fallback rate, mean over the group's sequences. k: every k-th frame processed.

	Base	ΔHOTA				CI of ΔHOTA			ΔIDF1				FB (%)		
Sequence	HOTA	Aff.	Tr.	ECC	Hyb.	Aff.	ECC	Hyb.	Aff.	Tr.	ECC	Hyb.	Aff.	Tr.	ECC
YOLOv8n detections															
05	43.4	+0.70	+0.73	+0.81	+0.66	[−0.40, 1.97]	[−0.17, 1.57]	[−0.49, 1.97]	+0.45	+0.69	+0.75	+0.28	1.7	7.3	0.0
10	32.0	+0.29	+0.20	+0.26	+0.29	[0.11, 0.64]	[0.08, 0.58]	[0.12, 0.64]	−0.03	−0.13	−0.10	−0.03	0.3	0.5	0.0
11	47.3	+0.29	+0.26	+0.29	+0.29	[0.14, 0.50]	[0.14, 0.51]	[0.14, 0.50]	+0.03	+0.02	+0.05	+0.03	0.0	1.3	0.0
13	27.8	+0.81	+0.15	+0.20	+0.91	[0.01, 1.26]	[−0.49, 0.87]	[0.11, 1.35]	+0.68	−0.07	−0.26	+0.71	2.9	28.8	0.0
Moving, $k = 1$	37.5	+0.48	+0.32	+0.38	+0.49	[0.18, 0.78]	[0.11, 0.61]	[0.18, 0.81]	+0.25	+0.08	+0.05	+0.22	1.2	9.5	0.0
Moving, $k = 2$	36.9	+1.07	+0.91	+1.37	+1.75	[0.33, 1.73]	[0.48, 2.26]	[0.86, 2.59]	+1.12	+1.00	+1.58	+2.12	8.7	34.3	0.0
Moving, $k = 3$	35.3	+2.37	+1.33	+3.97	+3.96	[1.24, 3.32]	[2.35, 5.21]	[2.33, 5.39]	+3.06	+1.79	+6.23	+5.92	19.2	67.0	0.0
Moving, $k = 6$	29.9	+2.02	+0.25	+5.43	+5.61	[1.20, 2.87]	[3.83, 6.83]	[3.96, 7.11]	+3.93	+0.66	+7.58	+7.55	61.8	92.7	0.2
Static, $k = 1$	33.5	−0.01	−0.01	0.00	−0.01		–		−0.03	−0.03	0.00	−0.03	0.0	0.0	0.0
Static, $k = 3$	35.9	−0.01	−0.01	−0.42	−0.01		–		−0.01	−0.01	−0.61	−0.01	0.0	0.0	0.0
MOT17 public FRCNN detections															
Moving, $k = 1$	47.9	+0.45	+0.36	+0.61	+0.58	[0.08, 0.83]	[0.22, 1.00]	[0.15, 1.00]	+0.92	+0.98	+1.13	+1.03	1.2	10.4	0.0
Moving, $k = 2$	45.5	+2.29	+1.13	+3.07	+3.33		–		+2.64	+1.30	+4.02	+4.06	10.6	37.0	0.0
Moving, $k = 3$	39.9	+4.15	+0.94	+7.03	+7.29		–		+5.40	+1.00	+9.55	+9.64	23.4	70.6	0.0
Moving, $k = 6$	31.6	+1.36	+0.02	+10.12	+9.83		–		+2.04	+0.10	+14.08	+13.63	67.6	94.1	0.2
Static, $k = 1$	47.5	+0.01	0.00	−0.01	+0.01		–		+0.01	0.00	−0.04	+0.01	0.0	0.0	0.0
Static, $k = 3$	49.1	+0.01	+0.01	−0.03	+0.01		–		+0.01	+0.01	−0.03	+0.01	0.0	0.0	0.0

Table S5: Synthetic motion on MOT17-02, -04, -09 (TrackEval combined over the three sequences). px: pan speed (px/frame) or shake σ (px). Δ_2, Δ_1: IDF1 change with CMC in the second (this table) and first run.

		IDF1			IDSW			ΔIDF1	
Motion	px	Base	CMC	Oracle	Base	CMC	Oracle	Δ_2	Δ_1
still	0	38.4	38.4	38.4	311	309	311	0.0	0.0
pan	2	37.9	37.9	38.3	380	380	382	0.0	+0.1
pan	5	38.9	40.1	40.1	399	403	403	+1.1	−0.8
pan	10	36.9	38.0	38.0	489	406	404	+1.1	+2.4
pan	20	27.8	40.2	40.2	987	371	369	+12.4	+14.3
shake	2	39.8	39.2	39.2	315	313	314	−0.6	0.0
shake	5	39.1	40.9	40.7	403	368	365	+1.8	−2.0
shake	10	38.3	39.5	39.5	649	353	355	+1.2	+4.8

Table S6: SportsMOT validation clips (first clip of each source video), YOLOv8n: baseline HOTA and Δ (estimator $-$ baseline).

Clip	Base	Aff.	Tr.	ECC	Hyb.	ΔIDF1 Aff.	IDSW
v-00HRwkvvjtQ-c001	20.4	−2.00	−1.47	−0.31	−1.68	−3.15	449
v-0kUtTtmLaJA-c004	27.5	+2.01	+2.37	+2.88	+2.01	+2.90	230
v-2QhNRucNC7E-c017	38.5	+3.34	−0.53	+3.46	+3.75	+3.71	188
v-4-EmEtrturE-c009	31.0	+1.76	+1.78	+2.93	+1.76	+1.89	87
v-4r8QL-wglzQ-c001	22.7	+0.93	+0.86	+1.28	+0.93	−0.30	293
v-5ekaksddqrc-c001	26.4	−0.17	−0.93	+0.03	−0.17	−0.14	183
v-9MHDmAMxO5I-c002	32.4	−1.04	−1.34	−1.13	−1.04	−1.90	94
v-BgwzTUxJaeU-c008	19.5	−0.12	+0.31	+0.30	−0.12	+0.70	185
v-G-vNjfx1GGc-c004	34.5	+2.82	−0.23	+0.63	+4.39	+5.93	199
v-ITo3sCnpw-k-c007	27.4	+3.93	+3.41	+3.26	+3.93	+4.01	388
v-cC2mHWqMcjk-c007	20.3	+0.41	+0.01	−0.90	+0.41	+0.62	342
v-dw7LOz17Omg-c053	27.4	+5.38	+3.33	+0.72	+5.38	+7.39	358
v-i2-L4qquVg0-c006	35.4	−0.47	+0.39	+0.23	−0.28	−1.37	290

Table S7: Estimator and tracker parameters (all fixed before the MOT17 runs except the hybrid rule)

Component	Setting
Corners	Shi–Tomasi, $\leq$1,000, quality 0.01, min. distance 8 px, block 3, half resolution
Mask	detection boxes of frame $t-1$
LK flow	window 21×21, 3 pyramid levels, OpenCV default termination (30 iter., $\epsilon = 0.01$)
RANSAC	`estimateAffinePartial2D`, 3 px, OpenCV defaults (2,000 iter., conf. 0.99)
Gates	$\geq$30 tracked points, inlier ratio $\geq$0.5; else identity
ECC	Euclidean, 10% scale, 100 iter., $\epsilon = 10^{-5}$, no mask; identity if not converged
DeepSORT	cosine 0.2, budget 100, IoU 0.7, max. age 30, $n_{\text{init}} = 3$, conf. $\geq$0.25

Table S8: CMC from PTZ angle readings: ΔHOTA and ΔIDSW against the baseline for reading noise σ (°) and delay (frames), PTZ sweeps on MOT17-02, -04, -09. Simulated readings; one run.

		ΔHOTA, delay			ΔIDSW, delay		
Sweep	σ	0	1	3	0	1	3
20 °/s	0	+2.28	+1.74	+0.66	−51	−41	+22
	0.1	+2.31	+1.88	+0.72	−46	−24	+68
	0.3	+0.95	+0.60	−0.75	+155	+221	+359
	image-based affine: ΔHOTA +1.80, ΔIDSW −58						
40 °/s	0	+5.32	+3.32	−0.98	−343	−152	+185
	0.1	+5.22	+3.57	−1.25	−335	−152	+222
	0.3	+3.85	+1.91	−2.66	−184	+89	+395
	image-based affine: ΔHOTA +5.38, ΔIDSW −337						

Table S9: Per-frame processing time on two Xeon cores at 2.10 GHz (CPU only; first run). All rows measured on this machine with `perf_counter` around each call; estimation includes grayscale conversion and resizing.

Component	ms/frame, median [IQR]
DeepSORT update, baseline	2.92 [1.98, 4.23]
DeepSORT update + state warp	3.20 [2.22, 4.60]
CMC estimation, 1920×1080	11.40 [10.54, 12.54]
CMC estimation, 640×480	2.01 [1.75, 2.29]
YOLOv8n detection, 1920×1080†	91.8
YOLOv8n detection, 640×480†	88.3
Appearance encoding, 1920×1080†	82.4
Appearance encoding, 640×480†	56.0

†Same machine and run; median over sequences of per-sequence medians, recorded during detection; depends on detections per frame.

Table S10: Evidence per claim. M: measured and reproduced; S: measured once; I: inconclusive (interval includes zero); U: not reproduced; E: exploratory; R: reported elsewhere; N: not tested.

Claim	Type	Basis
Little effect on static cameras	M	Table I, suppl.
Small HOTA gain on MOT17 moving sequences	M	Sec. V-E, suppl.
Gain not driven by one MOT17 sequence	S	leave-one-out
Full-rate IDF1 gain on MOT17, YOLOv8n	I	CI includes zero
Full-rate IDF1 gain on MOT17, FRCNN	S	CI excludes zero
Large gains with frame skipping	S	per-step Kalman only, Table S11
Gain grows with PTZ sweep speed (two detectors, seven scenes)	S	Table I
Estimation matches oracle except at very large per-frame motion	S	Table I
Small gain on a real sweeping camera	S	Table II
Small gain at 0.5° pole sway	S	Table I
PTZ-angle CMC needs low delay	S	Table S8, simulated
Fewer duplicate PPE reports at fast sweeps	E	Table III, simulated
Loss on forward-moving sequences	S	YOLOv8n only, Table IV
Loss at moderate synthetic motion	U	supplementary
Hybrid best or near-best estimator	E	not on new sequences
CMC cost on this CPU stack	M	Table S9
CMC gains in other trackers	R	Section II
Transfer to workers on real PTZ footage	N	future work

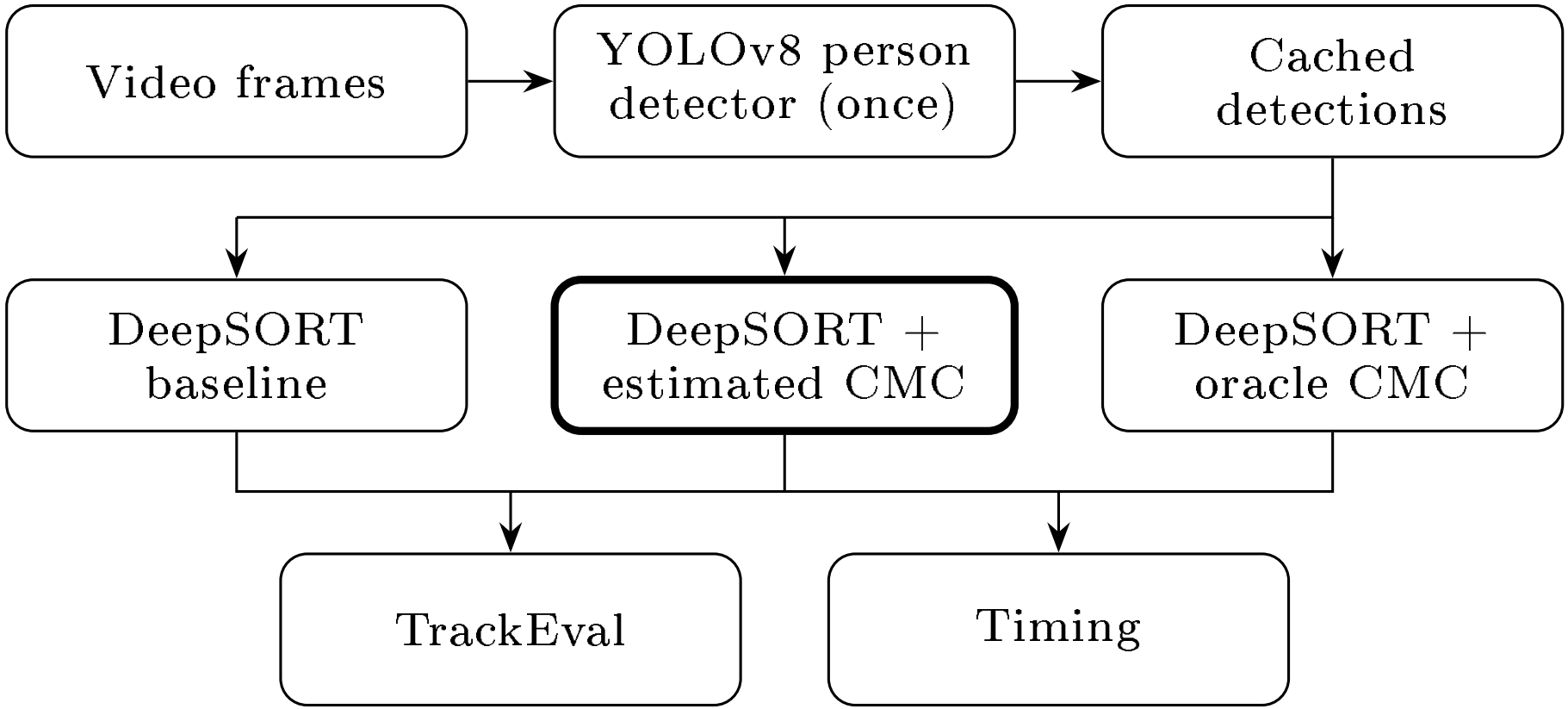


Figure S2: Experimental pipeline. All conditions receive identical cached detections and appearance features; only the CMC step differs. The estimated-CMC condition also reads the raw frames; the oracle condition (synthetic sequences only) uses the known camera offset. Figure produced by the authors.

Table S11: Frame skipping on moving sequences under the per-step (main) and time-aware Kalman models: baseline HOTA and ΔHOTA (condition − baseline), TrackEval combined. Hyb.: exploratory. Intervals for the time-aware model are given in the text.

		Per-step (main)				Time-aware			
Det.	k	Base	Aff.	ECC	Hyb.	Base	Aff.	ECC	Hyb.
YOLOv8n	2	36.9	+1.07	+1.37	+1.75	37.6	+0.18	+0.58	+0.50
	3	35.3	+2.37	+3.97	+3.96	38.7	+0.25	+0.94	+0.76
	6	29.9	+2.02	+5.43	+5.61	35.0	−0.74	+0.69	+0.56
FRCNN	2	45.5	+2.29	+3.07	+3.33	46.8	+0.65	+1.23	+1.34
	3	39.9	+4.15	+7.03	+7.29	45.2	+0.49	+2.30	+1.87
	6	31.6	+1.36	+10.12	+9.83	40.3	+0.28	+2.74	+2.87